



Training School 01

Delft, NL, March 18th-21st 2025

Welcome!

Mathias Peirlinck



Co-funded by
the European Union



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Confederation

Federal Department of Economic Affairs,
Education and Research EAER
State Secretariat for Education,
Research and Innovation SERI



UK Research
and Innovation

What is



Virtual Twins as tools for personalised clinical care

The VITAL study aims to revolutionize cardiovascular healthcare with cutting-edge technology, bridging clinical data, wearable devices, and advanced computational modeling to create virtual human twins.

This groundbreaking study will contribute to a new era of personalized, sex-specific treatment strategies for complex cardiovascular conditions, transforming how we approach systemic diseases.



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The agenda

	Tuesday March 18th 2025	Wednesday March 19th 2025	Thursday March 20th 2025	Friday March 21st 2025
08.45-9.00	walk-in	walk-in	walk-in	free
08.45-9.45	Welcome and Introduction.	0D/1D circulation modeling	Alya framework	free
09.45-10.30	Physical units for physiology	0D/1D circulation modeling calibration techniques (deep learning)	Alya tutorial	Functional connectivity map overview
10.30-10.50	coffee break	coffee break + posters 1	coffee break + posters 2	coffee break
11.00-11.45	Physical laws and bond graphs	0D/1D circulation modeling calibration tutorial	Bayesian inference	Functional connectivity - coupling 0D blood flow with cellular models - coupling multiphysics
11.45-12.30	Physical laws and bond graphs	0D/1D circulation modeling calibration tutorial	Bayesian inference	
12.30-13.30	lunch break	lunch + POSTER SESSION 1	lunch + POSTER SESSION 2	lunch break
13.30-14.15	Simple bond graph models	Uncertainty quantification and sensitivity analysis	Ion channel model	Surrogate modeling
14.15-15.00	Simple bond graph models (focus on 0D arterial modeling)	Uncertainty quantification and sensitivity analysis	Model of myofilament mechanics	Surrogate modeling
15.00-15.20	coffee break	coffee break + posters 1	coffee break + posters 2	Closing
15.30-16.15	0D circulation model / tutorial w openCOR/circulatory autogen	Parameter uniqueness/identifiability tutorial	Cellular (cardiomyocyte / neuron) modeling w openCOR	Final drinks
16.15-17.00	0D circulation model / tutorial w openCOR/circulatory autogen	Parameter uniqueness/identifiability tutorial	Cellular (cardiomyocyte / neuron) modeling w openCOR	
evening	Social activity at Mooie Boules		Dinner at De Kurk	

lectures
tutorial / exercise session
networking

The lecturers



Peter Hunter
University of Auckland



Mathias Peirlinck
Delft University of Technology



Gonzalo Maso Talou
University of Auckland



Finbar Argus
University of Auckland



Alberto Zingaro
ELEM biotech



Lydia Aslanidou
EPFL



Beatrice Ghitti
University of Auckland



Nikos Stergiopoulos
EPFL



Dimitrios Lialios
ELEM biotech



Rogier Krijnen
Delft University of Technology

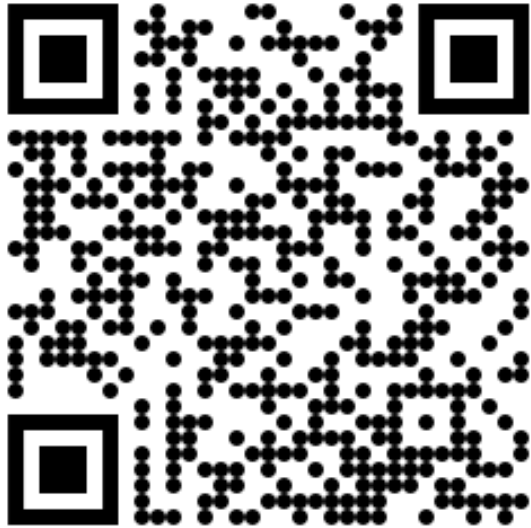


Wouter Huberts
Eindhoven University of Technology



Shauna O'Donovan
Eindhoven University of Technology

Training materials



When you use our training materials for your own research/education, please cite us.

zenodo

Posters

WEDNESDAY March 19th, 2025
Posters for participants with
surnames A – K on display

THURSDAY March 20th, 2025
Posters for participants with
surnames L – Z on display

*Please hang up your posters in
the morning and take them
down in the evening.*



TUESDAY 17h30-18h30



Mooie Boules Delft
Professor Schermerhornstraat 9, Delft

games + drinks/bites



Social events



Restaurant De Kurk Delft
Kromstraat 20, Delft

dinner



THURSDAY 19H00 - ...

Digital twins survey – we need your input!

Evaluation of knowledge and awareness regarding regulatory aspects of digital twins.

Aim: create tailored training based on identified gaps.



Please fill out:

*Regulation and medical devices:
a survey for the biomedical
modelling and digital twin
community*



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Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union, HADEA, SERI or UKRI. Neither the European Union nor the granting authorities can be held responsible for them.



Thank you